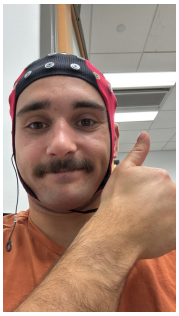

Piero Simonetto

Control System Engineer

Contact

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LinkedIn
GitHub

Address
Via Sibelius 59
Bassano del Grappa, VI
36061
Italy

Skill

Languages
Italian (Mother tongue), English (Fluent)

Lab
EEG-based MI Decoding, BCI for Human-Robot Interaction, Experimental Design Data Analysis

Software
Linux, LaTeX, ROS

Programming
C++, Matlab, Python3

Job related
Project Management Critical Thinking, Collaborative Organized, Creative Problem-Solving,

Personal
Cooking, Hiking, Wrestling

About Me:
Ph.D. student specializing in Human-Robot Interaction (HRI) and Brain-Computer Interfaces (BCI). My research focuses on integrating human brain activity with robotic systems to enhance interaction and autonomy. Passionate about teamwork, cutting-edge technology, and applied research in assistive robotics.

Education:

02/2023 - present, University of Padua Ph.D.: Information Engineering
Thesis: "Human-Machine Interfaces for Industry 5.0: Advancing Human-Robot Interaction through Real-Time Decoding of User Cognitive States."

09/2020 - 10/2022, University of Padua LM-25: Control System Engineering
Thesis: "Fault detection for boolean systems"

10/2017 - 09/2020, University of Padua L-8: Ingegneria dell'Informazione
Thesis: "Classical and Quantum entropy"

09/2012 – 07/2017, I.T.I.S. E. Fermi High school on Computer Science

Publications:

P. Simonetto, S. Toniolo, S. Tortora, E. Menegatti, L. Tonin
"Enhancing Motor Imagery Decoding with Environmental Context During Robot Control" - *2025 System Man & Cybernetics Conference (SMC)*

D. De Lazzari, P. Simonetto, N. Turcato, L. Tonin, R. Carli
"Nonlinear Model Predictive Control of a BMI-guided Wheelchair for Navigation in Unknown Environments" - *2024 European Control Conference (ECC)*

M. R. ur Rahman, P. Simonetto, A. Polato, F. Pasti, L. Tonin, S. Vascon
"OpenNav: Efficient Open Vocabulary 3D Object Detection for Smart Wheelchair Navigation" - *Twelfth International Workshop on Assistive Computer Vision and Robotics (ACVR)*

P. Simonetto, S. Tortora, E. Menegatti, L. Tonin
"Fusing user's intention and environment information in BMI driven robotic devices"
In Proceedings of 5th I-RIM Conference

Experience:

NEUROROBOTICS University of Padua
02/01/2023 - present, IASLAB at unipd
- Developing a BCI-driven navigation system for wheelchairs, ensuring user safety and comfort.
- Research supervised by L. Tonin.

NEUROROBOTICS UT at Austin, Ph.D. Mobility period
01/11/2024 - 04/30/2025, José del R. Millán lab
- Conducting real-time Error-Related Potential (ErrP) detection for powered wheelchair teleoperation.

WORKSHOP ORGANIZER WS5 at Summer School on Neurorehabilitation
08/06/2025 - 13/06/2025, SSNR
- Organized and managed a workshop on how to connect a BCI to a real device

TEACHING ASSISTANT University of Padua
02/10/2023 - 02/2/2024, Intelligence Robotics
- Manage the course laboratories, where the students learn how to use robots in a dynamic environment through ROS and C++

BCI Cybatlon BCI race - WHI team (2nd Position)
27/10/2024 , ETH Zurich
- Designed and optimized MI decoding strategies to enhance BCI performance in competition settings.

ERASMUS UCLouvain University
09/2021 - 01/2022, Louvain-la-Neuve
- Focused on Bioengineering Neurosciences research.

Internship Reflex&Allen
08/2016 - 09/2016, Wales, UK
- Assisted in company IT infrastructure management and troubleshooting.